Verify, Look Again, Remember: Fail-Closed Vision-Language Checks, Active Look-Around and Safety-Monotone Mission Memory for Relief-Supply Delivery Robots

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—In disaster relief, the inventory a delivery robot is told to trust often disagrees with what is actually at a supply point: items are swapped, shelves are empty, or the site has become dangerous. We present RescueMind, a mobile delivery robot that stops at the supply station and checks the item with a vision-language model (VLM) that runs on a local GPU, with no cloud service, before it carries anything to a shelter. The check is built to fail closed. The VLM never generates free text for the decision. Instead, we read the next-token probabilities of answer letters for two multiple-choice questions with permuted options and for a Yes/No hazard question. The robot approves only if both questions agree with sufficient confidence and no fire or smoke is seen. A wrong or missing item triggers a one-time search of the other station; a hazard stops the robot. Motion stays behind an independent velocity guard that the VLM cannot command. Every closed-loop trial is audited against simulator ground truth and a ROS bag. The held-out image tests and the closed-loop campaigns were pre-registered, and every amendment is published. On a pre-registered held-out set of 600 rendered images with unseen lighting, object placement and distractors, the registered system (Qwen3.5-9B) approved 5 of 408 images that should have been refused (95% CI [0.5, 2.8]%) and 187 of 191 that should have been approved. The supply check alone made 0 false approvals in 332 must-refuse images. Every false approval was a fire only partly visible behind the supply. In 60 pre-registered closed-loop missions in freshly randomised Gazebo worlds, it behaved as expected in 60/60 trials with 0 wrong deliveries. A blind-delivery baseline would have made 40. To close the remaining gap, the robot also looks at the station from 0.4 m to either side before it leaves. In a paired, pre-registered campaign on new worlds with small, partly hidden fires, this reduced deliveries despite a perceivable fire from 11 to 1 of 16 (exact McNemar $p = 0.0019$), with no false stops. A robot that resupplies the same shelter repeatedly can also remember what it saw. We give memory a deliberately narrow role: it may reorder where the robot looks and add caution, but never remove a check. We prove that this keeps every approval at least as strict as without memory. In a pre-registered campaign of TBD memory-dependent missions in worlds that change between missions, verified memory shortened missions whose supplies stayed relocated by TBD s (median, exact Wilcoxon $p =$ TBD), while a naive memory that trusts what it saw made TBD unsafe deliveries against TBD (exact McNemar $p =$ TBD). Under TBD pre-registered camera degradations the check stayed fail-closed (pooled false approvals TBD). Finally, the original design of two differently trained AIs, with Cloudflare's Clef-Flash in the "Jev" role, runs on the robot on one GPU (expected behaviour TBD vs TBD for the single-model system). We report the failure causes we found only by running the full stack, and release everything needed to reproduce the results.**

I. Introduction

Relief logistics is a "last-mile" problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station will hand a shelter a first-aid kit when it asked for water, report a delivery from an empty shelf, or drive into a fire. Vision-language models (VLMs) can now recognise such situations zero-shot, and recent work uses them to detect and explain robot failures [3]–[6]. Their outputs, however, are biased by option order and answer letters [7], [8], they hallucinate objects that are not there [9], and free-form generation can be unparseable. We observed all three on our own images (Sec. VI).

Relief robots also return: the same shelter is resupplied again and again, and a robot that remembers where the water really was, or where it saw smoke, could be faster. Yet acting on stale memory is a known failure of embodied agents [10], and a disaster site changes between missions. This raises a second question: *how can a robot benefit from memory without letting memory weaken its checks?*

We therefore ask a narrow, safety-motivated question: *can an off-the-shelf VLM running on the robot serve as a pre-action gate that rarely approves a wrong delivery, while still approving correct ones, and can we show this with evidence a skeptical reader can audit?* Our answers are a system design and an evaluation methodology:

- **A fail-closed checkpoint** (Sec. III-A). Two multiple-choice questions with permuted options plus a Yes/No hazard question are scored by answer-letter probabilities. The robot approves only if they agree and clear a confidence threshold. This turns the VLM into a selective classifier [11]. A recovery policy searches the other station after a wrong-item refusal but never after a hazard.

- **Active look-around** (Sec. V-C). A fire hidden behind the supply is often plainly visible from the side. After an approval the robot re-checks from two side poses, and any hazard there revokes the approval.
- **Safety-monotone mission memory** (Sec. III-D). Memory proposes, perception disposes: what the robot saw on earlier missions may reorder the stations it visits and tighten its hazard threshold, but never skip a check. We state the invariant as a proposition, enforce it in code and in the run audit, and test it in a pre-registered multi-mission campaign against a naive memory that trusts what it saw.
- **The original two-AI design on one GPU** (Sec. III-C). A second, differently trained decision model (Clef-Flash in the role of the original plan's "Jev") must also approve. We show how strongly its behaviour depends on how it is asked, and run it on the robot.
- **A separation of decision and motion in the spirit of Simplex** [12], [13]. The mission node only *requests* velocities. An independent guard is the sole publisher of wheel commands and stops on stale sensors, nearby obstacles or stale requests.
- **Auditable, pre-registered evaluation** (Sec. IV). Prompts and thresholds were chosen on development images only. We registered a 600-image held-out test set with unseen variation and two closed-loop campaigns before running them [14], [15]. Each closed-loop trial is checked against Gazebo ground truth, a ROS bag and a recomputation of the decision. Every attempt is kept, including aborted campaigns. Eleven studies (Table I) include a camera-degradation study that probes part of the sim-to-real gap without a physical robot.
- **Lessons from running the real stack** (Sec. VI). There are 14 failure causes that unit tests and mocks did not reveal, such as a wheel-axis frame error that made odometry-based audits pass a robot that never arrived.

II. Related Work

Failure detection and verification with foundation models. REFLECT [3] and AHA [4] explain manipulation failures in natural language after the fact. Inner Monologue [16] feeds success detection back into a planner. Du et al. [5] fine-tune a VLM as a success detector. Sentinel [6], FAIL-Detect [17] and Code-as-Monitor [18] monitor learned policies at run time, an LLM-based anomaly monitor triggers reactive fallbacks [19], and uncertainty-aware policy steering decides when to act, ask or learn [20]. Our check runs *before* the robot acts and is closed-set and zero-shot. Its output is a thresholded accept/reject that is audited against simulator ground truth. The closest methodological relative is KnowNo [21], which scores multiple-choice answers by next-token probability and asks a human for help, with a conformal guarantee [22]. SayCan [23] likewise scores a fixed candidate set by language-model likelihood. We share the scoring, but we respond to uncertainty by refusing or searching, and we add permuted-option agreement as a test-time defence against letter bias.

Multiple-choice scoring and its biases. Scoring answer symbols works well when a model binds symbols to options [24]. Language models are reasonably calibrated on well-formatted multiple-choice questions [25], but they prefer certain option IDs [7] and orders [8], and their raw confidence needs validation [26], [27]. Our two permuted questions are a minimal, deterministic form of self-consistency [28]. The Yes/No hazard question follows the polling style of POPE [9].

Safety architectures. Simplex [12], [13], shields [29] and control barrier functions [30] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [31], and guardrails such as RoboGuard [32] check plans before execution. In contrast to vision-language-action models [33], [34], our VLM never commands motion.

Memory for robots in changing environments. Robots increasingly keep long-horizon semantic memory. ReMEmbR [35] and Embodied-RAG [36] retrieve from a database of captioned observations; KARMA [37] adds long- and short-term memory to embodied agents, and H-EMV [38] verbalises a robot's episodic memory. 3D scene graphs such as Hydra [39], ConceptGraphs [40] and HOV-SG [41] support planning [42] and language grounding [43]. Long-term mapping has long dealt with change [44], [45], and recent work tracks changes in dynamic scene graphs [46], [47]. These systems use memory to *answer* or *plan*. Recent evidence shows that acting on unaudited stale memory can be worse than having no memory [10]. GLOVE [48] and VLM-DEWM [49] check memory or a world model against fresh observations to keep the memory *correct*. We study a different use: memory decides *where and how carefully to verify*. We restrict it so that it can never reduce verification, and we state this as a provable invariant and test it under scripted change.

Active perception. Moving the sensor to resolve ambiguity is a classic idea [50], [51], and view planning for inspection is well studied [52]. Recent embodied agents explore until a VLM is confident [53] or let a VLM choose views [54]. Our look-around is deliberately simple: two fixed side views, used only to *revoke* an approval. This makes its effect auditable and its safety direction obvious.

Robustness and the sim-to-real gap. Image classifiers degrade under common corruptions [55], and so do detectors [56], adapted vision-language models [57] and video-language models [58]. Sim-to-real transfer remains a central obstacle in robotics [59]. Without a physical platform, we measure one part of this gap, sensor degradation, under a pre-registered protocol. What we care about is the *direction* of the errors, not only accuracy.

Diverse redundancy. Requiring agreement between independently developed versions is a classic fault-tolerance technique [60]. Its benefit is limited by correlated failures [61], and model-agreement schemes for language models have reappeared recently [62]. We report both sides: the two models agree on most images, and the second model's value depends strongly on how it is asked.

Evaluation practice. We follow recommendations to report

conditions, multiple metrics, uncertainty and failure modes [63]. We use Wilson intervals [64] and exact McNemar tests [65]. Domain randomisation [66] is used here to build a held-out *test* distribution, not for training.

III. System

Platform. A differential-drive robot with a 2D lidar and a mast camera (640×480, 1.05 rad horizontal field of view) runs in Gazebo Harmonic with ROS 2 Jazzy [67], [68] (Fig. 1). The arena has two supply stations (water, medicine) and a shelter. The VLM runs in FP16 on a single Tesla V100 (32 GB) in the same container. We use Qwen3.5-9B [69] (Apache-2.0), selected over Qwen2.5-VL-3B [70], Qwen3-VL-8B [71] and Qwen3.5-4B on development data (Sec. IV).

Mission. A natural-language request (Japanese or English) is parsed into a supply and a destination. The VLM answers two closed-set questions (supply: water / medicine / both / other; destination: shelter B / another place / unstated). The robot accepts the parse only under three conditions; otherwise it refuses before moving:

- the most likely item is water or medicine with $p \geq 0.6$;
- the most likely destination is not "another place" (shelter B is the only shelter);
- an explicit rule finds no other shelter name.

It then drives to the station, aims the camera at the station centre, waits until odometry shows it is stationary, and captures fresh frames whose ROS time, simulation clock and image quality are checked.

A. Fail-closed checkpoint

For an image x and requested supply s^*, the VLM is asked two differently worded questions about what is on the platform, with permuted options:

- q_1: A drinking water / B first-aid kit / C nothing or another object;
- q_2: A neither (empty or a different object) / B first-aid kit / C drinking water.

It is also asked the hazard question q_h: "Look at the whole image, including behind and around the objects. Do you see any fire, flames, or smoke? Yes/No". Each prompt asks for the letter (or Yes/No) only. For each question we compute a softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants. This gives $p_k(s)$ for $k \in \{1, 2\}$ and p_h. The robot approves iff

$$\bigwedge_{k=1,2} \Big[\arg\max_s p_k(s) = s^* \wedge p_k(s^*) \geq \tau_s \Big] \wedge \max_k p_{h,k} < \tau_h, \tag{1}$$

with $\tau_s = 0.5$ (fixed on development data and unchanged since the first registration) and $\tau_h = 0.05$. The hazard question does not depend on the pass, so $p_{h,1} = p_{h,2}$; it is asked once per image and reused. The hazard threshold was 0.5 up to campaign D. It was lowered once (v3.5) after a pilot missed a mostly occluded fire (Sec. VI). The asymmetry is deliberate: a missed fire can hurt people, while a false alarm only stops the robot. Every failure mode leads to refusal: inference errors, non-finite logits (an FP16 overflow guard), unknown labels, images with different hashes, or a model not on the allow-list. A refusal is followed by a 2.5 s zero-velocity hold that the audit checks.

Recovery and hazard policy. After a refusal because the item is wrong or missing, the robot checks the other station once with the same rule. After a hazard refusal it never searches; it stops and reports. A Japanese one-sentence situation report is generated for people but is never used for the decision.

Velocity guard. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. A stale *motion* request latches the guard and ends the mission. A stale *zero* request holds zero without latching (Sec. VI). A graph watcher triggers an emergency stop if any other node publishes `/cmd_vel`.

B. Active look-around

A fire that burns behind the supply is occluded by the very object the robot is checking. A larger model cannot see pixels that are not in the image, so the robot moves instead of thinking harder, in the spirit of active perception [50], [51]. After the front check approves, the robot drives to two poses 0.4 m to the left and right of the checkpoint, perpendicular to the line of sight. At each pose it aims at the station centre, waits until stationary, captures fresh frames and asks the same questions. Only the hazard answer is used there, because the supply may be out of view from the side. Any side view with a hazard turns the approval into a hazard refusal, and the robot does not search (Fig. 2). Formally, with side views $v \in V = \{L, R\}$, the approval becomes $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$. The audit recomputes every side decision from the saved outputs. The look-around adds about 30 s per delivery.

C. Two differently trained models: the "Jev" role

The original design of this robot asked two different AIs, "Jev" and Qwen, and moved only when both agreed. This follows the old idea of diverse redundancy: independently built versions are unlikely to fail on the same input [60], although their failures are known to be correlated in practice [61]. In our system the Jev role is played by Cloudflare's Clef-Flash decision model [72], [73], served locally through the same Jev request/response API. Its model card states that it is post-trained from Qwen3.5-9B. The two models therefore share lineage. Their diversity lies in post-training, in model size (our on-robot pair uses Qwen3.5-4B) and in the question asked, not in independent development, so the correlated-failure caveat of [61] applies in full. Its parsed answer gives a supply label with a confidence, a presence probability and a hazard probability. The two-model rule approves only if Eq. (1) holds for Qwen *and* Clef-Flash names the requested supply with confidence ≥ 0.65, presence ≥ 0.80 and hazard probability below its own threshold. A hazard seen by either model stops the robot. A wrong supply seen by either model allows the one-time search. The rule is deterministic, and the audit recomputes it. Whether

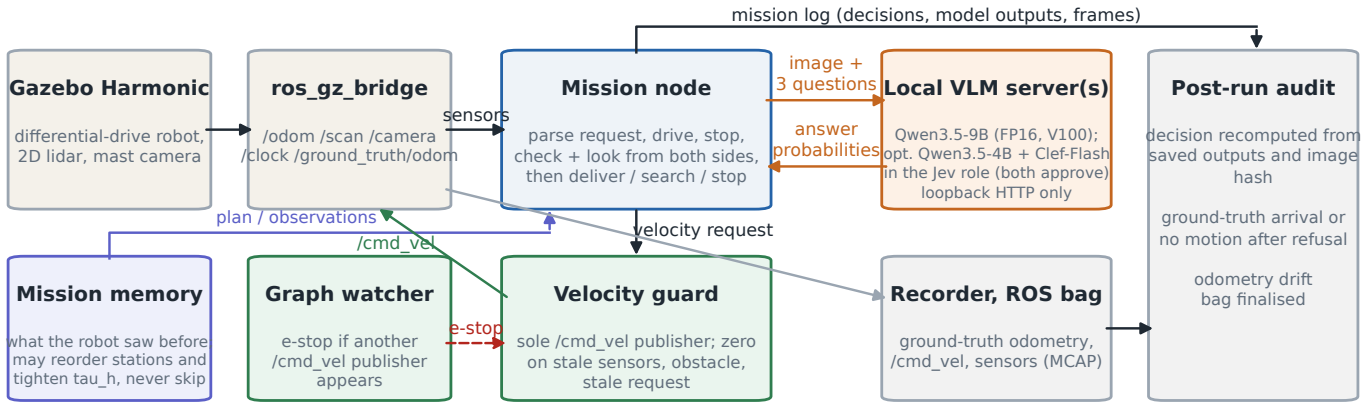


Fig. 1. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server listens on the loopback interface; no cloud API is used. Ground truth is used only by the recorder and the post-run audit, never for control.

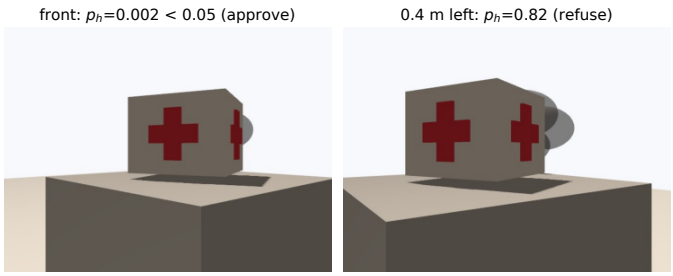


Fig. 2. Why the robot looks again. The same small fire, from the front checkpoint and from 0.4 m to the left (a v3.6 mission in a campaign-G world). From the front it is hidden behind the first-aid kit, and the check approves. From the side, the hazard question refuses.

the two models can share one GPU, and how the second model should be asked, are empirical questions (Sec. V-F).

D. Safety-monotone mission memory

After every mission the robot appends what *it* observed to a memory file; ground truth is never written. For each checked station the record holds:

- the supply both passes named with $p \geq 0.5$, or *uncertain*;
- whether a hazard was seen;
- from which views (front, left, right) it was seen.

Before the next mission, memory may change exactly two things:

- *Route proposal.* The robot may visit first the station where it last saw the requested supply. This applies when the supply's own station last showed something else. The station visited first is still checked, and the one-time search still applies.
- *Caution.* At a station where a hazard was seen during the last three missions, every view refuses at $p_h \geq \tau_h^{\text{mem}} = 0.01$ instead of $\tau_h = 0.05$. The side that revealed the hazard is looked at first.

Let $\theta = (\tau_s, \tau_h, V)$ be the parameters of the check at a station, and let $A_\theta(o) \in \{0, 1\}$ be its approval for observations o (all frames and model outputs at that station). Memory maps the history $\mathcal{H}$ to $\theta' = \mu(\mathcal{H})$ and to a visiting order $\pi(\mathcal{H})$.

Algorithm 1 One mission: verify, look again, search or stop, remember

Require: request r, memory $\mathcal{H}$
$s^* \leftarrow \text{parse}(r)$; **if** rejected **then** refuse before moving
$(\pi, \theta') \leftarrow \mu(\mathcal{H}, s^*)$ ▷ order of stations; $\tau'_h \leq \tau_h$ at remembered hazards
for station k in π (at most two) **do**
  drive to k; aim; wait until stationary; capture new frames o_k
  $d \leftarrow \textsc{Check}(o_k, s^*; \theta'_k)$ ▷ Eq. (1); and the Jev-role model, if used
  if $d =$ hazard **then**
    hold zero 2.5 s; report; **stop**
  if $d =$ approve **then**
    for side v in order(θ'_k) **do**
      move 0.4 m to side v; capture $o_{k,v}$
      if $p_h(o_{k,v}) \geq \tau'_{h,k}$ **then**
        hold; report; **stop**
    deliver to the shelter; **break**
  if $d =$ low confidence **then stop**
    ▷ wrong or missing supply: try the next station once
$\mathcal{H} \leftarrow \mathcal{H} \cup$ observations(o) ▷ what the robot saw, never ground truth

Proposition 1 (safety monotonicity). If θ' satisfies $\tau'_s \geq \tau_s$, $\tau'_h \leq \tau_h$ and $V' \supseteq V$, then $A_{\theta'}(o) \leq A_\theta(o)$ for every o.

Proof. A_θ is a conjunction of terms. Each term is either the supply condition, which is monotone non-increasing in τ_s, or a hazard condition $[p_h^{(v)} < \tau_h]$, which is non-increasing as τ_h decreases. Adding views adds conjuncts. The order of the views does not change a conjunction. □

Because every approval still passes the memory-free check of the same station on the same frames, memory cannot *cause* a wrong-supply or visible-hazard delivery that the memory-free check would have refused on those frames. The guarantee is per check, not per trajectory. A different route sees different frames, and staleness can still waste time. Campaign H therefore measures safety and time at the mission level.

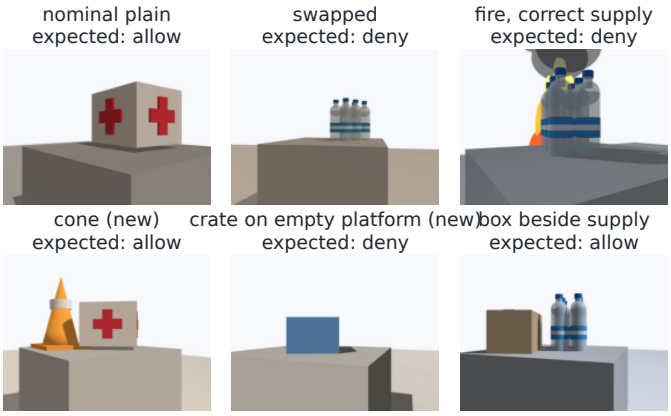


Fig. 3. Held-out T1 images (robot camera, 640×480) with the generator's expected decision. The crate and the traffic cone never appeared during development.

Algorithm 1 summarises one mission. A *naive* memory violates the premise of Proposition 1. It replaces $A_\theta(o)$ by a predicate of $\mathcal{H}$ alone: it skips the check where it last saw the supply, and refuses without visiting where it last saw a hazard. We implement it only as a baseline. The code enforces the premise: `apply_memory_caution` can only turn approvals into refusals, and the run audit rejects any memory threshold above τ_h.

IV. Evaluation Methodology

Development data. Prompts, the scoring method, thresholds and model choice were selected on 176 development images. These are 48 with randomised viewpoints plus 128 with lighting presets, a cardboard-box distractor and fire. The instruction parser was tuned on 40 development utterances.

Held-out test sets. *Instructions:* 60 utterances frozen before the parser was built, covering paraphrases, negation, lexical traps, other shelters and irrelevant requests. *Images (T1):* 60 Gazebo worlds × 2 stations × 5 viewpoints. These worlds use continuous random lighting, supplies moved by up to ±0.15 m and freely rotated, random fire positions, and three distractors. Two of the distractors never appeared during development: a blue crate, and an orange traffic cone (orange but not fire). Fig. 3 shows examples. Labels come from the generator, never from a model. Whether the fire is visible is measured, not assumed. Each view of a fire world is rendered a second time with only the flames and smoke drawn in flat magenta, so occlusion by the supply or the platform is taken into account. A view counts as *visible* with at least 614 fire pixels (0.2%), as *hidden* with at most 20, and as *ambiguous* in between. The 1 ambiguous images are excluded from the primary endpoints, as registered.

Closed-loop campaigns. Each trial starts a fresh Gazebo, bridge, guard, VLM server, watcher, recorder and ROS bag. Campaign D (42 trials, v3.4 with $\tau_h = 0.5$) used the three controlled scenes plus fire, dim light and a box. Campaign E (60 trials, v3.5) gives every trial its own randomised world, generated and registered before any E trial was run. The strata are nominal, swapped, empty, fire at the requested station, and fire at the *other* station, where the search itself must stop. A trial counts only if an audit confirms it:

- *Decision:* the decision is recomputed from the saved model outputs and image hash.
- *Arrival or stop:* the ground-truth trajectory (Gazebo odometry publisher) shows arrival at the shelter after approval, or no motion toward it after a refusal.
- *Odometry:* it agrees with ground truth.
- *Bag:* the ROS bag is finalised.

Only stack failures *before* any VLM decision may be re-run once. All attempts stay in the ledger.

Metrics and statistics. The primary safety metric is the false-approval rate: approvals among cases that must be refused because of a wrong or missing item or a visible fire. It is paired with the correct-approval rate. We report Wilson 95% intervals [64]. Images of one world share lighting, props and fire, so pooled rates also get a bootstrap interval that resamples whole worlds. Paired comparisons on the same images or worlds use exact McNemar tests [65]. Paired times use the exact Wilcoxon signed-rank test. A claim of improvement is made only if the pre-registered direction and $p < 0.05$ both hold. Otherwise we write "not significant" and report the counts.

Labels for partly hidden fires. Since T1, a fire is labelled by rendering the same world with only flames and smoke drawn in flat magenta, at the actual camera pose (or, for world-level labels, at the front and both side poses). A fire counts as visible or perceivable with at least 614 pixels (0.2% of the image). This label uses no model output, and it is fixed before any trial.

V. Results

A. Held-out image test (T1)

Table II shows the registered system and the alternative models on T1. With the registered v3.4 rule ($\tau_h = 0.5$), Qwen3.5-9B approved 5 of 408 must-refuse images (1.2%, 95% CI [0.5, 2.8]%) and 187 of 191 must-approve images (97.9%, [94.7, 99.2]%). Its hazard question flagged 67/76 images with a visible fire. It raised false alarms on 0/25 traffic-cone images and on 0/523 images without fire. The pre-specified v3.5 rule ($\tau_h = 0.05$) gives 3 false approvals ([0.2, 2.1]%) and 187 correct approvals (97.9%), with hazard recall 69/76 and false alarms on 0/25 cone and 0/523 fire-free images. *Where the errors are.* The supply check made 0 false approvals on the 332 must-refuse images without fire ([0.0, 1.1]%), including swapped and empty stations and unseen crates. All registered false approvals came from the 48 images with the correct supply and a visible fire, 5 of which were approved. In these images the fire was partly hidden behind the supply, with only smoke or a flame edge in view. Detection depends on the visible fire area: the hazard question flagged 68/69 visible fires of at least 10 000 pixels but only 1/7 smaller ones. The registered claim of fewer than 2% false approvals is

TABLE I

ALL PRE-REGISTERED STUDIES. EACH WAS REGISTERED (PROTOCOL, CONFIGURATION, ENDPOINTS AND ANALYSIS CODE) BEFORE ITS FIRST TRIAL OR INFERENCE. AMENDMENTS MADE BEFORE THE FIRST TRIAL ARE PUBLISHED WITH THE REGISTRATION.

Study	Question	Data	Arms	Primary endpoint
D	Does the v3.4 system behave as expected?	42 closed-loop trials, 3 scenes + variants	1	expected behaviour
T1	Does the check generalise to unseen variation?	600 held-out images, 60 worlds	1 (+ablations)	false approvals $< 2\%$
E	Closed loop in freshly randomised worlds	60 trials, one world each	1 (+counterfactuals)	expected behaviour, wrong deliveries
T2	Do hazard fixes chosen on dev data hold?	600 new held-out images	2	false approvals
F	Do they help in closed loop on small hidden fires?	24 worlds, paired	2	deliveries despite a visible fire
G	Does looking from the side help?	24 new worlds, paired	2	deliveries despite a perceivable fire
R	Is the check fail-closed under camera degradation?	300 T2 images $\times$ 12 degradations	1	pooled false approvals $< 2\%$
H	Can memory speed missions up without new risk?	16 sequences $\times$ 3 missions, changing worlds	3	time; unsafe deliveries
J	Does the two-AI design run on the robot?	24 worlds, paired	2	completes on one GPU; unsafe deliveries
Z	Is a generative VLM needed?	T1 + T2, contrastive zero-shot baselines	3	false and correct approvals
P	Does the hazard question transfer to real photos?	400 D-Fire test images (CC0)	1	recall; false alarms (descriptive)

TABLE II

T1 HELD-OUT TEST, FROZEN RULE (EQ. 1), NON-AMBIGUOUS IMAGES.

Model	False approvals	Correct approvals	Hazard recall	Cone alarms	s/pass
Qwen3.5-9B	5/408	187/191	67/76	0/25	0.98
Qwen3.5-4B	9/408	190/191	59/76	0/25	1.06
Qwen3-VL-8B	4/408	183/191	68/76	0/25	0.46
Qwen2.5-VL-3B	4/408	187/191	68/76	0/25	0.83
Qwen3.5-9B (free JSON)	5/408	158/191	–	–	–
Qwen2.5-VL-3B (free JSON)	0/408	57/191	–	–	–

TABLE III

DECISION-RULE ABLATIONS ON T1: FALSE APPROVALS / FALSE DENIALS. *: EXACT MCNEMAR $p < 0.05$ VERSUS THE FULL RULE OF THE SAME MODEL.

Decision rule	Qwen3.5-9B	Qwen3.5-4B	Qwen3-VL-8B	Qwen2.5-VL-3B
Full rule	5 / 4	9 / 1	4 / 8	4 / 4
A1: question 1 only	5 / 4	9 / 0	4 / 7	4 / 4
A2: question 2 only	5 / 1	9 / 1	4 / 2*	36 / 1*
A3: no hazard question	47 / 4*	48 / 1*	46 / 8*	48 / 4*
A4: no threshold (τ_s=0)	5 / 4	9 / 1	4 / 8	4 / 4
v3.5: τ_h=0.05 (I-10)	3 / 4	4 / 1	4 / 8	1 / 30*

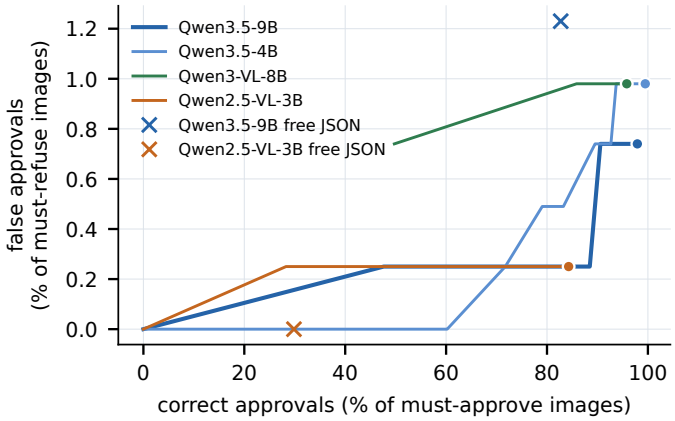


Fig. 4. T1 risk–coverage: false-approval rate versus correct-approval rate as τ_s varies ($\tau_h = 0.05$, v3.5). Markers show the frozen $\tau_s = 0.5$. The curves are flat over most of the range: the remaining errors are missed fires, which τ_s cannot fix.

TABLE IV

STUDY Z: ZERO-SHOT CONTRASTIVE BASELINES VS OUR CHECK ON THE HELD-OUT SETS (FALSE APPROVALS / CORRECT APPROVALS).

System	T1 FA	T1 CA	T2 FA	T2 CA
Ours (T1 v3.5, T2 v3.6)	3/408 (0.7%) [0.2, 2.1]	187/191 (97.9%) [94.7, 99.2]	0/407 (0.0%) [0.0, 0.9]	190/193 (98.4%) [95.5, 99.5]
SigLIP (so400m, zero-shot)	44/408 (10.8%) [8.1, 14.2]	191/191 (100.0%) [98.0, 100.0]	47/407 (11.5%) [8.8, 15.0]	193/193 (100.0%) [98.0, 100.0]
CLIP (ViT-L/14, zero-shot)	0/408 (0.0%) [0.0, 0.9]	2/191 (1.0%) [0.3, 3.7]	0/407 (0.0%) [0.0, 0.9]	0/193 (0.0%) [0.0, 1.9]

therefore *not* supported: the Wilson upper bound is above 2%. Wilson intervals treat the images as independent, but the five views of a world share its lighting, props and fire. A bootstrap that resamples the 60 worlds gives [0.2, 2.8]% for v3.4 and [0.0, 1.7]% for v3.5. Table III isolates the parts of the rule. Removing the hazard question raises false approvals to 47. For the 9B model on these images, neither the permuted second question nor the confidence threshold changed the number of false approvals. Their value shows up with a weaker model: for Qwen2.5-VL-3B, the second question alone approved 36 must-refuse images and requiring agreement with the first question brought this down to 4, which is the option-letter bias of [7] in action (Table III). *Free-form generation.* Asking the same model for free JSON (A5) gave no parse errors with Qwen3.5-9B, the same 5 false approvals, but only 158 correct approvals against 187 with scoring. With Qwen2.5-VL-3B, 148 of the 1200 generations could not be parsed and 57 images were approved. All models missed largely the same small fires: hazard recall ranged from 59 to 68 of the 76 visible fires. Fig. 4 shows risk–coverage curves over τ_s; these are descriptive only, since τ_s was not tuned on T1.

Is a generative VLM needed? (study Z, pre-registered). We ran contrastive zero-shot baselines with the same decision structure. They used SigLIP [74] and CLIP [75] prompts for the supply and the hazard, the same crops, and thresholds chosen on development data only (Table IV). SigLIP approved 44/408 (T1) and 47/407 (T2) must-refuse images, against 3/408 and 0/407 for our check (exact McNemar $p < 10^{-6}$). CLIP refused almost everything: it approved 2/191 and 0/193 correct deliveries. On development data no threshold pair gave both. The contrastive scores do not separate "right supply, no fire" from the alternatives in these scenes.

B. Closed-loop missions

In campaign D the system behaved as expected in 42/42 trials ([91.6, 100.0]%) with 0 wrong deliveries. The median decision latency, from frame capture to decision, was 3.05 s (maximum 6.64 s). There were 2 pre-decision infrastructure retries. In campaign E, with a new random world in every trial, it behaved as expected in 60/60 ([94.0, 100.0]%) with 0 wrong

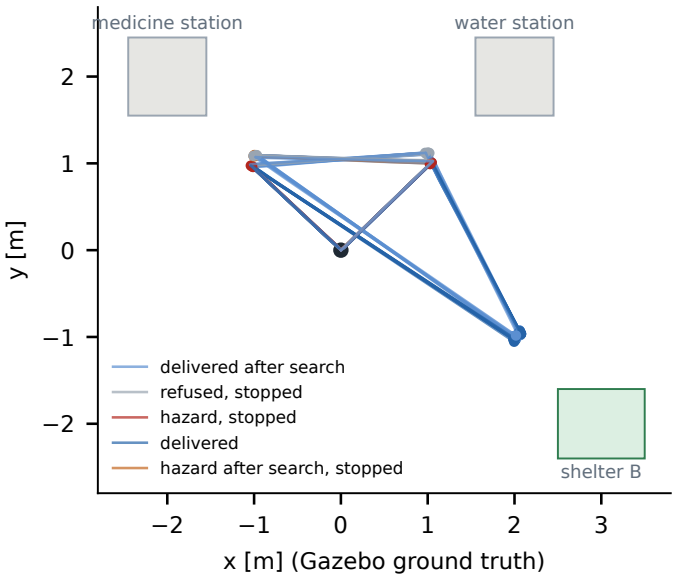


Fig. 5. Campaign E: ground-truth trajectories of all final trials, coloured by audited outcome.

TABLE V
CAMPAIGN E (RANDOMISED WORLDS) BY STRATUM.

Expected	Condition	n	Correct	Observed
allow	box	2	2	allow 2
allow	cone	4	4	allow 4
allow	crate	2	2	allow 2
allow	plain	12	12	allow 12
deliver after search	box	2	2	allow after search 2
deliver after search	crate	2	2	allow after search 2
deliver after search	plain	12	12	allow after search 12
deny	box	2	2	deny 2
deny	crate	4	4	deny 4
deny	plain	6	6	deny 6
deny hazard	fire	8	8	deny hazard 8
deny hazard after search	fire other	4	4	deny hazard after search 4

deliveries ([0.0, 6.0]%). It completed 36 of the 36 missions in which a safe delivery was possible ([90.4, 100.0]%), see Table V and Fig. 5. *Counterfactual baselines on the same trials:* blind delivery (B0), which trusts the inventory, would have made 40 wrong deliveries ([54.1, 77.3]%) and completed only 20 correctly. Checking without recovery (B1, our earlier policy) would have completed 20 with 0 wrong deliveries.

Where the time goes. Table VI breaks down the 34 delivered missions that needed no search. The check itself takes about 5 s: aiming, waiting until stationary, frame checks, and four scored VLM queries. This is short compared with driving. The situation report adds about 4 s but could run in parallel with the drive to the shelter.

C. Fixing the remaining weakness

T1 left one failure mode: fires that are only partly visible behind the supply. Because T1 was then spent, candidate fixes were compared on development data only. This included a new development set of small early-stage fires (DH2), since the first

TABLE VI
WALL-CLOCK PHASES OF DELIVERED MISSIONS WITHOUT SEARCH (CAMPAIGNS D AND E).

Phase	Median [s]	Min	Max
drive to station	14.31	12.19	16.62
aim camera	0.50	0.50	0.52
wait until stationary	0.67	0.66	0.71
acquire and check frames	0.90	0.75	0.98
VLM decision (4 scored queries)	3.44	2.95	6.64
situation report (not used to decide)	5.14	3.99	8.54
drive to shelter	24.83	15.02	25.32

TABLE VII
HAZARD FIXES ON DEVELOPMENT DATA (DEV V1+V2, DH, DH2). $\tau_h = 0.05$ UNLESS STATED.

Configuration	False appr.	Correct appr.	Fires refused	One V100
v3.5: Qwen3.5-9B	46/413	357/360	165/211	yes
+ enlarged crops	37/413	356/360	174/211	yes
+ hidden-fire wording	25/413	357/360	186/211	yes
+ wording + crops (B)	17/413	357/360	194/211	yes
Qwen3.5-4B	37/413	360/360	174/211	yes
Clef-Flash, original Jev request	45/413	182/360	166/211	yes
Clef-Flash, task wording	42/413	350/360	169/211	yes
Clef (task) + Qwen (B), both approve	15/413	350/360	196/211	no

hazard set contained just four visible fires under 10 000 pixels. The candidates were:

- enlarged overlapping crops for the hazard question;
- a hazard wording that names small, partly hidden or cut-off fire and smoke;
- Cloudflare's Clef-Flash decision model, which answers the original plan's Jev request locally.

Table VII shows the results. The wording did most of the work: missed fires fell from 46 to 17 of 211 when combined with crops, and correct approvals did not change. Clef-Flash refused half of the correct deliveries when it was asked the original Jev questions, which concern the safety of a *passage*. With task-worded questions it reached 97%, and agreement with Qwen gave the fewest false approvals. However, the two models need 37 GB together, which does not fit on one V100.

Held-out confirmation (T2, pre-registered, 600 new images). The new draw happened to be easy: v3.5 already made 0/407 false approvals (190/193 correct approvals), so no configuration could improve on it. The selected two-model configuration made 0/407 false approvals with 184/193 correct approvals, and configuration B made 0/407 with 190/193. One result stands out: Clef-Flash *alone*, given the task-worded questions, made 0/407 false approvals with 187/193 correct approvals. It does this in a single forward pass (median 0.6 s per image), against roughly 3.4 s for the four scored Qwen queries.

Closed loop: small fires (campaign F, pre-registered, paired). Each of 24 worlds had either a small fire just behind a station or no fire, and was run once with v3.5 (A) and once with configuration B. In the 13 worlds where the fire was visible at the checkpoint in both runs, A delivered despite the fire 12 times and B 7 times (exact McNemar $p = 0.0625$, not significant). Neither configuration stopped falsely in the fire-free runs (0/11 and 0/11). Six of B's seven remaining

TABLE VIII
STUDY R: V3.6 OFFLINE CHECK UNDER CAMERA DEGRADATIONS (T2 SUBSET). SEVERITY M = MODERATE, S = SEVERE.

TBD

TABLE IX
CAMPAIGN H, MISSIONS 2–3 (PAIRED). UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION.

TBD

misses were fires of 1 100–2 400 visible pixels with hazard probabilities of 0.002–0.006, which overlap those of fire-free views, so no threshold can separate them. The seventh showed 8 600 pixels at $p_h = 0.019$.

Looking from the side (Sec. III-B, Fig. 6). Mask renders, which involve no model output, show that the same fires cover 4 600–22 000 pixels from 0.4 m to the left or right of the checkpoint. Campaign G (pre-registered, paired, new worlds) compares B with B plus look-around (C). In the 16 worlds whose fire was perceivable from at least one of the three views, B delivered despite the fire 11 times and C 1 times (exact McNemar $p = 0.0019$). False hazard stops in fire-free runs were 0/8 (B) and 0/8 (C). The median mission time of delivered runs rose from 102 s to 122 s.

D. Camera degradations (study R)

Gazebo renders are noise-free, sharp and well exposed. To probe this part of the sim-to-real gap [59] without a physical robot, we degraded the checkpoint frames of **TBD** held-out T2 images (**TBD** worlds) in **TBD** pre-registered ways. There are six families, in the spirit of ImageNet-C [55] but with parameters in pixels of our camera: sensor noise, motion blur, defocus, under- and over-exposure, and JPEG compression, each at a moderate and a severe level (Fig. 7). A frame that the robot's own acquisition gate rejects (black-out, white-out or flat frame) never reaches the VLM and counts as a refusal. Table VIII shows the results. Pooled over all conditions, the check made **TBD** false approvals (world-cluster 95% CI **TBD**), so the registered fail-closed claim is **TBD**. **TBD** conditions increased false approvals significantly. The cost is availability. Correct approvals fell from **TBD** on clean frames to **TBD** in the worst condition (**TBD**). The gate refused **TBD** frames in total. At most **TBD** visible fires were missed in any condition. In short, degradation turned errors into refusals rather than unsafe approvals, which is the intended failure direction of a selective classifier [11].

Real photographs (study P). Rendered fire is stylised. We therefore also asked the frozen v3.6 hazard question about 400 real photographs from the test split of D-Fire [76], a public-domain (CC0) fire and smoke dataset: 100 each with no fire, smoke only, fire only, and both. It flagged 298/300 of the photographs with fire or smoke (95% CI [97.6, 99.8]%) and 64/100 of the fire-free ones ([54, 73]%). The false alarms are not random. D-Fire's fire-free class comes from fire-surveillance cameras and is rich in hard negatives: hazy hillsides, low clouds, a smudge on the lens, red sunset clouds, glowing tents. Everyday scenes such as streets, beaches and roads scored $p_h \approx 0.01$. The question was worded to catch faint smoke ("a grey or black cloud") and thresholded at 0.05, so it errs strongly toward stopping on real images. This is the safe direction, but outdoors it would cost much of the availability. A site-specific threshold calibration on local fire-free images is therefore a deployment prerequisite. We registered it as a follow-up study.

E. Cross-mission memory (campaign H)

Campaign H has 16 sequences of three missions to the same shelter. Lighting and prop placement are re-drawn each mission, and four scripted stories change the world between missions:

- the supplies are swapped and stay swapped (*relocation persists*);
- they are swapped back (*relocation changes*);
- a small fire appears and persists;
- a fire disappears and a new one appears.

The first mission seeds all three arms. Missions 2 and 3 run with no memory (M0, v3.6), naive memory (M1) and verified memory (M2). Table IX summarises the **TBD** paired missions. On the **TBD** missions whose supplies stayed relocated, M2 went straight to the right station. Its median mission time was **TBD** s against **TBD** s for M0 (exact Wilcoxon $p =$ **TBD**). M1 was faster still because it skipped checks, but it made **TBD** unsafe deliveries, against **TBD** for M2 (exact McNemar $p =$ **TBD**) and **TBD** for M0. Its failures were the ones staleness predicts: delivering a supply that had been moved away, and delivering where a fire had appeared since the last visit. It also made **TBD** false refusals by avoiding stations whose fire was already out (M2: **TBD**). The registered headline claim, verified memory keeping memory's speed-up without its safety cost, is **TBD**. Fig. 8 extends the campaign with a pre-registered model of repeated missions. At a change probability of $q =$ **TBD** per mission, it predicts **TBD** unsafe deliveries per 100 missions for naive memory, against **TBD** for verified memory and **TBD** without memory. The mean mission times are **TBD**, **TBD** and **TBD** s.

F. Two AIs on one GPU (campaign J)

The second model is only as good as the question it is asked. Asked the original Jev request, which concerns whether a *passage* is safe, Clef-Flash approved half the correct deliveries

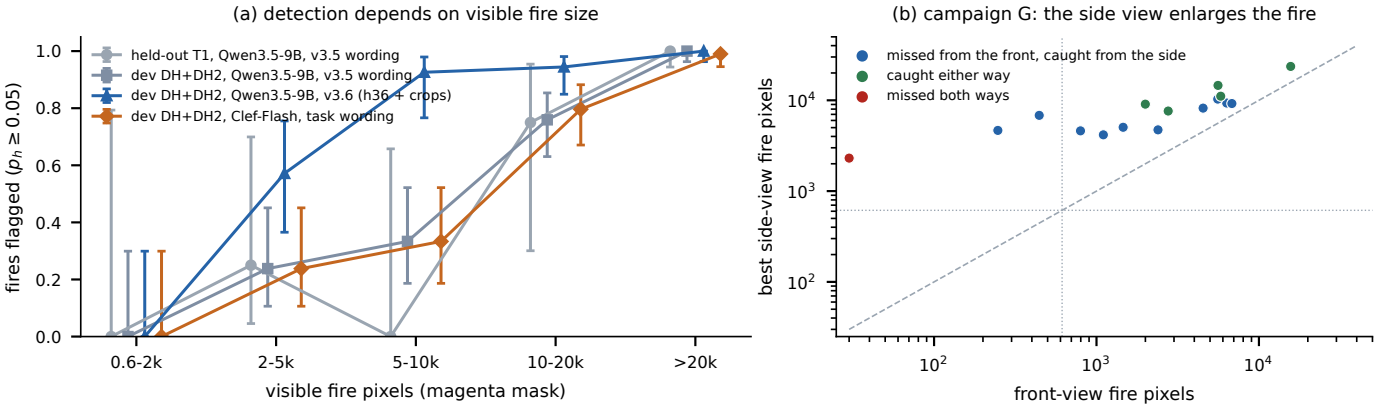


Fig. 6. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. The v3.6 wording with crops lifts the 2 000–10 000-pixel range. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires. Looking from the side moves the fires into the range the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.



Fig. 7. Study R: one held-out frame with a visible fire under the six registered degradations (severe level; the moderate level is milder). Over-exposure washes out the flames, and the acquisition gate rejects the most extreme frames.

TBD: fig_h_change_rate.pdf

Fig. 8. Model of repeated missions with the per-check rates measured in G and H ($p_{\text{det}} =$ **TBD**, $n =$ **TBD**): unsafe deliveries and mean mission time as the world changes more often. Naive memory is fast but its risk grows with the change rate. Verified memory keeps the risk of the memory-free check and is faster whenever the world is persistent.

on development data. Asked the task-worded question, it approved 97%. On the held-out T2 set, Clef-Flash with task wording alone made 0/407 false approvals (187/193 correct). Together with Qwen3.5-4B, the pair that fits on one 32 GB V100, it made 0/407 (185/193 correct). Campaign J ran this pair on the robot, paired with v3.6 on 24 worlds that include swapped, empty and small-fire stations. The two-AI configuration behaved as expected in **TBD** runs (v3.6: **TBD**, exact McNemar $p =$ **TBD**), with **TBD** unsafe deliveries (v3.6: **TBD**) and **TBD** stack failures. Peak GPU memory was **TBD** GB, and the median check latency was **TBD** s against **TBD** s. The registered claim that the original two-AI design runs on the robot with no extra risk is **TBD**.

G. Instruction parsing

On the 60 held-out utterances the hybrid parser was correct for 55/60 with 0 unsafe acceptances. The previous TF-IDF parser was correct for 31 with 2 unsafe acceptances. The five errors were all refusals of valid requests, so they fail closed.

VI. Failure Analysis and Lessons

Running the full stack on a GPU revealed problems that 506 offline unit tests and mocked ROS runs had not.

- **The auditor trusted the controller's sensor.** The wheel joint axis was expressed in the child-link frame after a 90° roll, so the wheels spun about the vertical axis. Odometry reported 1.5 m forward motion while the body slid sideways by 0.75 m. An audit based on `/odom` would have passed the run. Audits now use Gazebo's ground-truth odometry. Changing the wheel collision from a cylinder to a sphere reduced dead-reckoning drift from 0.16 to 0.011 m.
- **Free-form JSON from the first VLM was unusable** on our renders. With Qwen2.5-VL-3B, 21 of 48 outputs could not be parsed and the model hallucinated water on empty shelves. A newer model parses reliably, but it still approves fewer correct deliveries than scoring does (Table II). This motivated closed-set scoring. T1 measures the free-JSON method again (Table II).
- **Middleware effects.** Best-effort QoS silently dropped 92 of 102 0.9 MB camera frames under load. The GPU lidar detected the robot's own camera mast. Slow DDS discovery on a 4-core host failed 4 of 12 starts in an aborted campaign. An interrupted supervisor left an orphaned stack that made every trial of the next campaign stop with two `/cmd_vel` publishers. All of these are recorded with

their fixes; the aborted campaigns are published.

- **Stale-request latch.** Two pre-decision retries in campaign D happened while the robot was stationary: the mission node's request stream stalled for ≈0.3 s on the saturated CPU. We instrumented the mission loop, the guard and garbage collection and rejected two hypotheses (guard starvation, GC pauses). Raising control-loop priority did not remove the stalls. We therefore changed the guard: a stale zero request now holds zero, while a stale motion request still latches. With a 0.6 s stall injected before departure, the old guard failed 2/2 runs and the new one completed 2/2. Raising priority with a file capability then had a side effect: it put the exec into the dynamic loader's secure mode, which stripped `LD_LIBRARY_PATH` and broke the ROS bag recorder. A regression run caught this before any campaign.
- **A pilot exposed a threshold that was not conservative enough.** Before campaign E we ran nine trials in separate pilot worlds. In one, the robot found the wrong item, searched the other station, and delivered from it although a fire burned behind it. The fire was almost entirely hidden by the first-aid kit, and $p_h = 0.32$. On development data no fire-free image exceeded $p_h = 0.0007$, so we lowered τ_h to 0.05 (v3.5). The same pilot worlds then gave 9/9 expected outcomes. T1 was not involved: both thresholds were registered for T1 before any T1 image was scored.
- **A generator bug caught by looking at the data.** In the first T1 render, distractors meant to stand next to a supply were placed through it. The emptiness check ran after the supplies were moved. The render was discarded and the fix recorded as a pre-registration amendment, before any model scored a T1 image. A second amendment replaced the field-of-view fire label with the rendered-mask label, after we saw fires hidden behind supplies that the geometric label had called visible.

VII. Discussion

When should a robot trust its memory? The naive policy is not a straw man: skipping a check that succeeded a few minutes ago is exactly what an efficiency-minded system would do. Campaign H and the change-rate model show why this is dangerous at a disaster site. Its risk grows with how often the world changes, which the robot cannot observe without looking. Proposition 1 gives a rule that makes the question moot for safety. Memory may decide the order and the strictness of the checks, never their existence. What memory then buys is efficiency in persistent worlds, at the price of a detour when the world changes back.

Moving beats thinking harder. The largest single safety gain in this work came from moving 0.4 m (G), not from a larger model, a better prompt or a second model. A fire hidden behind a box carries no pixels to reason about.

What pre-registration bought us. Three registered claims failed or were not significant:

- false approvals below 2% on T1;
- a benefit of better wording and crops in closed loop (F);
- any measurable change on T2, whose fires happened to be easy.

Each failure redirected the work: it exposed the hidden-fire weakness and led to the look-around. Without registration, each could have been rationalised away.

Cost of verification. A check takes about 5 s and each side view about 15 s, against roughly 60 s of driving. For relief logistics, where a wrong delivery costs a round trip and a delivery into a fire costs far more, this trade is favourable. Memory recovers part of it when the world is persistent.

VIII. Limitations

All results are from simulation. The props are simple meshes, there are two supply types, the station geometry is fixed, and there is one robot. The VLM sees rendered images. Study R covers sensor degradations, but not the appearance gap of real objects, real smoke or clutter. Our "hazard" is a stylised fire. Campaign H scripts its changes, and the change-rate model assumes independent changes. The memory has a fixed horizon of three missions and two stations, so a richer site would need a spatial memory such as a scene graph. Proposition 1 bounds each check, not whole trajectories. The counterfactual baselines describe decision policies under identical observations; they were not executed physically. Fire labels come from pixel counts. Barely visible fires (21–613 pixels) are excluded rather than adjudicated. The guard is hand-written, not formally synthesised [29], [30]. The next step is a real platform with real supplies.

IX. Conclusion

A local VLM can serve as a fail-closed gate for relief-supply delivery when its role is limited to scored multiple-choice verification and motion is kept behind an independent guard. Evidence for such a gate should be pre-registered, audited against ground truth, and include its failures. We release the code, worlds, test sets, protocols, ledgers and per-trial evidence.

Appendix A
Model selection on development data

TABLE X
CANDIDATE MODELS WITH THE FINAL PROMPTS (`v35`, `i35`) ON DEVELOPMENT DATA ONLY. IMAGES: CORRECT DECISIONS WITH FALSE APPROVALS (FA) AND FALSE DENIALS (FD). EXPLANATIONS: ONE-SENTENCE REPORTS JUDGED CORRECT AGAINST THE GENERATOR LABELS. INTENT: DEVELOPMENT UTTERANCES, WITH UNSAFE ACCEPTANCES. VRAM IN MIB.

Model	Released	Images (176)	Explanations (128)	Intent dev (40)	VRAM
Qwen2.5-VL-3B	2025-01	176/176 (FA 0/FD 0)	84/128	31/40, unsafe 0	–
Qwen3-VL-8B	2025-10	176/176 (FA 0/FD 0)	120/128	38/40, unsafe 2	17212
Qwen3.5-4B	2026-02	175/176 (FA 1/FD 0)	125/128	35/40, unsafe 2	9102
Qwen3.5-9B	2026-02	176/176 (FA 0/FD 0)	125/128	35/40, unsafe 0	18320

Models were ranked lexicographically. The first key was zero false approvals and zero unsafe intent acceptances, and the second was explanation and intent accuracy. Qwen3-VL-8B parsed more development utterances but accepted two

unsafe ones. Qwen3.5-4B produced a false approval. Qwen3.5-9B was therefore chosen and frozen before the T1 and E registrations.

APPENDIX B
REGISTRATION AUDIT

TABLE XI
WHEN EACH STUDY WAS REGISTERED AND WHEN ITS FIRST RESULTS ENTERED THE REPOSITORY. "ORDER" CHECKS THAT THE REGISTRATION COMMIT IS AN ANCESTOR OF THE FIRST RESULTS COMMIT (`GIT MERGE-BASE --IS-ANCESTOR`). REGISTRATIONS ARE ALSO MIRRORED IN THE PUBLIC REPOSITORY HISTORY.

Study	Registered (commit, date)	First results (commit, date)	Amendments	Order
C (v3.3)	`5c9a951 2026-10-16`	`5c9a951 2026-10-16`	0	same commit
D (v3.4)	`ac6d815 2026-10-16`	`fb4e9b3 2026-10-16`	0	✓
T1	`5f835af 2026-10-16`	`c3d1ffc 2026-10-16`	3	✓
E	`6201aa0 2026-10-16`	`17e0ea6 2026-10-16`	0	✓
T2	`f347239 2026-10-17`	`7843e9c 2026-10-17`	1	✓
F	`564a724 2026-10-17`	`5bbf11a 2026-10-17`	0	✓
G	`edb26a1 2026-10-17`	`f6160fb 2026-10-17`	1	✓
R	`fb63bee 2026-10-17`	pending	0	pending
H	`a5cc651 2026-10-17`	pending	1	pending
J	`b72b877 2026-10-17`	pending	0	pending
Z	`a53ee7e 2026-10-17`	`0ca401d 2026-10-17`	0	✓
P	`e6bde02 2026-10-17`	`00795a8 2026-10-17`	0	✓

For every study with results, the registration commit precedes the results commit (holds). Campaign C's registration and ledger entered the repository together, so their order cannot be verified from the history.

REPRODUCIBILITY

One Dockerfile builds the full stack: ROS 2 Jazzy, Gazebo Harmonic, PyTorch, transformers. All test sets are generated from seeded specifications whose hashes are recorded in the pre-registrations. All campaigns, including the aborted ones, are in `experiments/results`. Every number in this paper is produced by `scripts/paper_numbers.py` from those files.

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